Fundamental Modules - 1

WEEK 1: THE VEHICLE & ITS HEART

Days 1-6 | Detailed Hour-by-Hour Structure

Week Theme: *"Understanding what makes a vehicle move + Building basic mechanical literacy"*

DAY 1: VEHICLE OVERVIEW + WORKSHOP SAFETY + VEHICLE LIFTING

Learning Objectives:

- Understand complete vehicle architecture and major subsystems
- Master workshop safety protocols and PPE usage
- Perform safe vehicle lifting and support procedures
- Identify major structural components on actual vehicles

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Workshop Safety Fundamentals

Workshop Safety Culture:

- Why safety comes before everything else
- Common workshop accidents and how to prevent them
- Safety as a professional responsibility
- Shop safety rules and discipline

Personal Protective Equipment (PPE):

- Safety shoes: steel toe requirements
- Safety glasses: when to wear (always!)
- Gloves: when to use, when NOT to use (rotating machinery)
- Coveralls/work clothes: proper workshop attire
- Hearing protection for noisy operations

Workshop Hazards:

- Slip, trip, and fall hazards
- Chemical hazards: oils, cleaners, battery acid
- Fire hazards: fuel, oil-soaked rags, electrical
- Electrical hazards: battery, charging system
- Moving vehicle hazards: never work under unsupported vehicle
- Exhaust fumes: carbon monoxide danger

Emergency Procedures:

- Fire extinguisher locations and types (ABC)
- First aid kit location
- Emergency exits
- Spill cleanup procedures
- Accident reporting

Hour 2 (10:00 AM - 11:00 AM): Vehicle Architecture & Systems Overview

Vehicle Structure Basics:

- Unibody vs body-on-frame construction
- Major structural elements: A-pillar, B-pillar, C-pillar
- Firewall: separating engine from passenger compartment
- Floor pan, rocker panels, cross members
- Front and rear subframes
- Crumple zones and safety cell concept

Major Vehicle Systems (Survey Level):

- **Powertrain:** Engine, transmission, drivetrain
- **Chassis:** Suspension, steering, brakes, wheels/tires
- **Electrical:** Battery, starting, charging, lighting
- **Body:** Doors, panels, glass, interior
- **Safety:** Airbags, seat belts, ABS
- **Comfort:** HVAC, audio, power accessories

Vehicle Classifications:

- Body types: sedan, hatchback, SUV, MUV
- Drive types: FWD, RWD, AWD, 4WD
- Fuel types: petrol, diesel, CNG, electric
- Transmission types: manual, automatic, AMT, CVT

Hour 3 (11:00 AM - 12:00 PM): Vehicle Lifting Safety Theory

Why Lifting Safety Matters:

- Gravity never sleeps: vehicle weight and danger
- Real accidents from improper lifting
- Professional responsibility to self and others

Lifting Equipment Types:

- Floor jack: hydraulic operation principle
- Jack stands: load ratings and safety factors
- Vehicle lift/hoist: two-post vs four-post
- Wheel ramps: when and how to use

Jack Point Identification:

- Manufacturer-designated lift points
- Frame rails and reinforced areas
- Where NOT to place jack (body panels, suspension parts)
- Different jack points for different vehicles
- Using service manual to find correct points

Safe Lifting Procedure:

1. Vehicle on level surface, parking brake applied
2. Chock wheels that remain on ground
3. Locate correct jack point
4. Position jack perpendicular to vehicle
5. Lift slowly while watching for shifting
6. Position jack stands under designated points
7. Lower vehicle onto stands before working
8. Never rely on jack alone - always use stands
9. Test stability before going under vehicle

Critical Safety Rules:

- Never go under vehicle supported only by jack
- Always use jack stands rated for vehicle weight
- Never use bricks, wood blocks, or makeshift supports
- Check stand stability before working underneath
- Never start engine with vehicle on stands (unless instructed)

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Vehicle Identification Laboratory

Activity 1: Body Component Identification (30 min)

- Walk around vehicle exterior
- Identify and touch each component:
  - A-pillar, B-pillar, C-pillar (on sedan/SUV)
  - Firewall (visible from engine bay)
  - Rocker panel (under doors)
  - Wheel arches and fenders
  - Hood, trunk, doors
- Understand structural vs cosmetic components
- Take photos and label for reference

Activity 2: Under-Hood Component Mapping (30 min)

- Open hood safely (locate release, support rod/strut)
- Identify major components:
  - Engine (count cylinders if visible)
  - Battery (locate positive/negative terminals)
  - Radiator and cooling fan
  - Air filter housing
  - Brake fluid reservoir
  - Windshield washer reservoir
- Understand "component packaging" - how everything fits
- Close hood properly

Pass Criteria:

- ✅ Can identify all major body structural elements
- ✅ Can open/close hood safely
- ✅ Can locate 6+ major under-hood components

Hour 2 (2:00 PM - 3:00 PM): Safe Vehicle Lifting Practice

Activity 1: Floor Jack Operation (Instructor Demonstration - 20 min)

- Instructor demonstrates proper floor jack usage
- Shows correct jack point location

- Demonstrates lift procedure step-by-step
- Shows common mistakes (wrong jack points)
- Students observe and ask questions

Activity 2: Jack Stand Selection and Placement (20 min)

- Understanding load ratings (tonnage)
- Identifying jack stand contact points
- Proper stand height adjustment
- Positioning stands under vehicle safely
- Testing stability before trusting weight

Activity 3: Supervised Lifting Practice (20 min)

- Students work in pairs (one operates, one observes)
- Under close instructor supervision:
  1. Chock wheels
  2. Position floor jack correctly
  3. Lift vehicle slowly
  4. Place jack stands
  5. Lower onto stands
  6. Test stability
  7. Raise vehicle again
  8. Remove stands
  9. Lower completely

SAFETY NOTE: Each pair supervised individually by instructor. No simultaneous operations until competency verified.

Pass Criteria:

- ✅ Can locate correct jack points on vehicle
- ✅ Can operate floor jack safely
- ✅ Can position jack stands correctly
- ✅ Demonstrates understanding of safety sequence

Hour 3 (3:00 PM - 4:00 PM): Wheel Removal & Installation Practice

Activity 1: Wheel Fastener Identification (15 min)

- Lug nut vs lug bolt systems
- Identifying socket size needed

- Understanding left-hand vs right-hand threads (rare but exists)
- Wheel locks and special tools
- Torque specifications location (door jamb sticker)

Activity 2: Wheel Removal Procedure (20 min)

- Loosen lug nuts BEFORE lifting (1/4 turn only)
- Lift vehicle and secure on stands
- Remove lug nuts completely (counter-clockwise)
- Pull wheel straight off hub
- Inspect wheel mounting surface for rust/dirt
- Inspect brake components (visual only)

Activity 3: Wheel Installation Procedure (25 min)

- Clean hub surface if needed
- Align wheel with studs/holes
- Hand-thread lug nuts (start all before tightening)
- Tighten in star/cross pattern: WHY this matters
- Lower vehicle to ground
- Final torque in star pattern with torque wrench

Torque Pattern Demonstration:

```
4-lug pattern:     5-lug pattern:
    1                  1
  4   2              5     2
    3                  3   4
```

Pass Criteria:

- ✅ Can remove wheel safely
- ✅ Can install wheel with correct pattern
- ✅ Understands star pattern importance
- ✅ Can use torque wrench at basic level

DAY 1 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies correct PPE for workshop tasks
- [ ] Locates major vehicle structural components

- [ ] Locates correct jack points
- [ ] Operates floor jack safely
- [ ] Positions jack stands correctly
- [ ] Removes and installs wheel with proper sequence
- [ ] Uses star pattern for lug nuts

Knowledge Check (Verbal):

- Why must we use jack stands and not just a jack?
- What is the purpose of the firewall?
- Why do we tighten lug nuts in a star pattern?
- Where do you find the correct torque specification for lug nuts?

DAY 2: ENGINE BASICS PART 1 + BASIC HAND TOOLS

Learning Objectives:

- Understand 4-stroke engine operation cycle
- Identify major engine components
- Select and use basic hand tools correctly
- Practice safe tool handling procedures

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): The 4-Stroke Cycle

Introduction to Internal Combustion:

- What "internal combustion" means
- Difference from external combustion (steam engine)
- Chemical energy (fuel) → Heat energy → Mechanical energy
- Why we need 4 strokes to produce power

Stroke 1: INTAKE (Suck)

- Piston moves DOWN from TDC to BDC
- Intake valve OPEN, exhaust valve CLOSED
- Air-fuel mixture sucked into cylinder
- Crankshaft rotation: 0° to 180°

Stroke 2: COMPRESSION (Squeeze)

- Piston moves UP from BDC to TDC
- BOTH valves CLOSED
- Air-fuel mixture compressed (pressure increases)
- Compression ratio concept (e.g., 10:1)
- Crankshaft rotation: 180° to 360°

Stroke 3: POWER (Bang)

- Spark ignites mixture just before TDC
- Explosion forces piston DOWN
- BOTH valves remain CLOSED
- Only stroke that produces power
- Crankshaft rotation: 360° to 540°

Stroke 4: EXHAUST (Blow)

- Piston moves UP from BDC to TDC
- Exhaust valve OPEN, intake valve CLOSED
- Burned gases pushed out
- Crankshaft rotation: 540° to 720° (2 full rotations)

Understanding the Complete Cycle:

- 4 strokes = 2 crankshaft revolutions = 720°
- Only ONE power stroke per 4 strokes
- Why we need multiple cylinders
- Cylinder firing order concept

Combustion Principles:

- Fuel + Air + Spark = Power
- Stoichiometric ratio: 14.7:1 (air to fuel)
- Lean vs rich mixture
- Why timing of spark is critical
- Detonation vs normal combustion

Hour 2 (10:00 AM - 11:00 AM): Engine Construction & Components

Engine Block (Foundation):

- Cast iron or aluminum construction
- Cylinders: inline, V-configuration, flat/boxer

- Cylinder bore and stroke dimensions
- Water jackets for cooling
- Oil passages for lubrication
- Main bearing caps and crankshaft support

Cylinder Head (The Lid):

- Contains combustion chambers
- Valves: intake and exhaust (2 or 4 per cylinder)
- Valve springs and retainers
- Spark plug location
- Camshaft position (OHV vs OHC)

Crankshaft (Motion Converter):

- Converts linear piston motion to rotation
- Main journals: supported by block
- Rod journals: connect to pistons
- Counterweights for balance
- Flywheel attachment point

Pistons (The Movers):

- Piston crown: combustion pressure
- Piston skirt: cylinder wall contact
- Piston rings: compression and oil control
- Piston pin (wrist pin): connects to connecting rod

Connecting Rods (The Links):

- Small end: piston pin connection
- Big end: crankshaft connection
- Rod bearings at big end
- Transfers force from piston to crank

Camshaft (Valve Controller):

- Lobes control valve opening/closing
- Driven by crankshaft (belt or chain)
- Runs at 1/2 crankshaft speed (why?)
- Valve timing control

Valves (Traffic Control):

- Intake valve: lets air-fuel mixture in

- Exhaust valve: lets burned gases out
- Valve face and seat relationship
- Valve spring closes valve
- Valve clearance/lash concept

Hour 3 (11:00 AM - 12:00 PM): Basic Hand Tools Introduction

Spanners/Wrenches:

- **Open-end spanner:** Two open jaws, 15° angle
  - When to use: good access needed
  - Limitation: can slip off if worn
- **Ring spanner:** Closed loop, grips all 6 points
  - When to use: more torque needed, safer grip
  - Limitation: needs more clearance to swing
- **Combination spanner:** Open-end + ring
  - Most versatile tool in toolbox
  - Always match size to fastener

Sockets and Ratchets:

- **Socket drives:** 1/4" (small), 3/8" (medium), 1/2" (large)
- **Socket types:** 6-point (better) vs 12-point (more positions)
- **Ratchet mechanism:** Internal pawl allows reverse without removal
- **Direction switch:** Check before using!
- **Extensions:** Short, medium, long
- **Universal joint:** Angle access

Screwdrivers:

- **Flat/Slotted:** Oldest type, many sizes
- **Phillips:** Cross pattern, self-centering
- **Pozidriv:** Similar to Phillips, more grip
- **Torx:** Star pattern, very secure
- **Hex/Allen:** Internal hex socket
- Matching screwdriver to screw head is critical

Pliers:

- **Combination pliers:** General gripping
- **Long-nose pliers:** Reaching tight spaces

- **Diagonal cutters:** Wire cutting
- **Water pump pliers:** Adjustable jaw
- NOT for fasteners - will round them off

Tool Safety Rules:

- Use correct tool for job (never improvise)
- Check condition before use (worn = dangerous)
- Keep tools clean and organized
- Never use damaged tools
- Pull toward you, don't push (safer if slips)
- Use penetrating oil on stuck fasteners
- Never use cheater bar on ratchet (breaks mechanism)

Metric vs Imperial:

- India: Metric system (mm)
- Old vehicles: might have imperial (inches)
- Never use imperial tool on metric fastener or vice versa
- Common metric sizes: 8, 10, 12, 13, 14, 17, 19, 22mm

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Engine Component Identification

Activity 1: Display Engine Observation (30 min)

- Instructor shows cutaway/display engine
- Students identify each component:
  - Block with cylinders visible
  - Crankshaft in place
  - Pistons and connecting rods
  - Cylinder head
  - Camshaft and valves
  - Timing belt/chain
- Trace one cylinder's complete 4-stroke cycle
- Count total cylinders and configuration

Activity 2: Component Relationship Understanding (30 min)

- Draw simple engine diagram showing:
  - Piston-rod-crank connection

    - Camshaft-to-valve connection
    - Timing belt/chain connecting crank and cam
    - Intake and exhaust valve positions
- Understand why camshaft runs at 1/2 speed
- Visualize complete power flow

Pass Criteria:

- ✅ Can identify 8+ major engine components
- ✅ Can explain basic 4-stroke cycle
- ✅ Can trace piston-to-crankshaft connection

Hour 2 (2:00 PM - 3:00 PM): Basic Hand Tools Practice

Activity 1: Tool Identification & Selection (20 min)

- Layout of common tools on bench
- Identify each tool type and size
- Match tool to fastener:
    - Given bolt, select correct spanner
    - Given bolt, select correct socket
    - Understand "one size smaller = rounded bolt"
- Practice reading size markings on tools

Activity 2: Spanner & Ratchet Technique (20 min)

- **Proper grip:** Palm on handle, not fingertips
- **Pull, don't push:** Safer if tool slips
- **Ratchet direction:** Check before use
- **Socket engagement:** Fully seat socket on nut
- **Removing tight bolts:**
    - Try both directions first (might be left-hand thread)
    - Use penetrating oil if stuck
    - Tap gently with hammer to break corrosion
    - Never use excessive force - will round off

Activity 3: Practice Board Exercise (20 min)

- Practice board with various bolts/nuts installed
- Students practice:
    - Removing bolts with correct spanner

    - Using ratchet and socket
    - Changing ratchet direction
    - Using extensions
    - Installing bolts hand-tight
- Speed is NOT important - correct technique IS

Pass Criteria:

- ✅ Can select correct tool for given fastener
- ✅ Demonstrates proper grip and pulling technique
- ✅ Can operate ratchet with direction changes
- ✅ Uses correct socket size without damaging fastener

Hour 3 (3:00 PM - 4:00 PM): Simple Component Removal Practice

Activity 1: Air Filter Housing Removal (20 min)

- Locate air filter housing on vehicle
- Identify all fasteners holding it:
    - Screws (Phillips or Torx)
    - Clips or clamps
    - Hose connections
- Select correct tools
- Remove housing systematically:
    1. Loosen hose clamp
    2. Disconnect intake hose
    3. Remove mounting screws/clips
    4. Lift housing away
- Inspect air filter condition
- Reinstall in reverse order

Activity 2: Battery Hold-Down Removal (20 min)

- Locate battery in engine bay
- Identify hold-down bracket and fasteners
- Safety: Do NOT disconnect battery cables (not today's lesson)
- Remove hold-down:
    1. Identify fastener type (usually 10mm or 13mm)
    2. Select correct socket
    3. Remove fasteners completely
    4. Lift bracket away

5. Reinstall bracket and torque properly

- Understand "reverse sequence" for installation

Activity 3: Valve Cover Fastener Practice (20 min)

- Identify valve cover on engine
- Count number of bolts (typically 8-12)
- Identify bolt size needed
- Practice removal sequence:
  - Start from outside, work inward
  - OR start from center, work outward
  - Loosen in stages (don't remove one completely first)
- Remove 2-3 bolts only (not removing cover)
- Reinstall with proper sequence
- Understand why sequence matters (warpage prevention)

Pass Criteria:

- ✅ Can remove simple bolted components
- ✅ Understands "reverse sequence" for installation
- ✅ Keeps track of removed fasteners
- ✅ Uses correct tools without damaging fasteners

DAY 2 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies major engine components on display
- [ ] Selects correct hand tool for given task
- [ ] Demonstrates safe tool handling technique
- [ ] Removes simple bolted components
- [ ] Reinstalls components in correct sequence
- [ ] Organizes removed fasteners

Knowledge Check (Verbal):

- Name the 4 strokes in order
- Which stroke produces power?
- Why do we use a ring spanner instead of open-end for tight bolts?
- What does the crankshaft do?

DAY 3: ENGINE BASICS PART 2 + FASTENER IDENTIFICATION

Learning Objectives:

- Understand fuel system components and flow
- Understand ignition system components
- Master timing concepts
- Identify different fastener types and grades
- Understand proper tightening sequences

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Fuel System Fundamentals

Fuel System Overview:

- Purpose: Deliver correct amount of fuel to engine
- System components from tank to cylinder
- Petrol vs diesel differences (high-level)

Fuel Path Flow:

1. **Fuel Tank:** Stores fuel, has level sender
2. **Fuel Pump:** Electric pump (in tank or inline)
   - Creates pressure (2-6 bar for petrol)
   - Pumps continuously when engine running
3. **Fuel Filter:** Removes dirt and contamination
   - Critical for injector protection
   - Should be replaced per service schedule
4. **Fuel Lines:** High-pressure hoses/pipes
   - Fuel rail delivers to all injectors
5. **Fuel Injectors:** Spray fuel into engine
   - Electronic control by ECU
   - Open/close time controls fuel amount
6. **Pressure Regulator:** Maintains system pressure
   - Returns excess fuel to tank

Fuel Injection Types (Awareness Level):

- **Multi-Point Fuel Injection (MPFI):**
  - One injector per cylinder

    - Sprays into intake port
    - Most common system
- **Direct Injection (GDI/TGDI):**
    - Injector sprays directly into cylinder
    - Higher pressure system
    - Better efficiency
- **Throttle Body Injection (TBI):**
    - Single injector before throttle
    - Older technology

Fuel System Components to Identify:

- Fuel tank and filler neck
- Fuel pump (in tank)
- Fuel filter location
- Fuel lines (usually metal with rubber sections)
- Fuel rail on engine
- Individual injectors

Hour 2 (10:00 AM - 11:00 AM): Ignition System Principles

Ignition System Purpose:

- Create high voltage spark (up to 40,000V)
- Deliver spark at correct time
- Ignite air-fuel mixture in cylinder

Ignition System Components:

1. Battery (12V Source):

- Provides initial power
- Low voltage (12V)

2. Ignition Coil (Transformer):

- Converts 12V to high voltage
- Primary winding: 12V side
- Secondary winding: high voltage side
- Modern: one coil per cylinder (coil-on-plug)
- Older: single coil with distributor

3. Spark Plugs:

- Create spark across gap
- Gap typically 0.6-1.0mm
- Center electrode to ground electrode
- Heat range selection important
- Platinum/iridium last longer

4. Ignition Control Module/ECU:

- Electronic control of timing
- Determines when to fire spark
- Advances/retards timing based on:
  - Engine speed (RPM)
  - Engine load (throttle position)
  - Temperature

5. Crankshaft Position Sensor:

- Tells ECU when piston is at TDC
- Critical for timing calculation

Ignition Timing Concepts:

- **Spark must occur BEFORE TDC (during compression)**
- Typical: 10-15° BTDC at idle
- Advances at higher RPM (more time for flame to spread)
- Why: flame takes time to propagate
- Too early = detonation/knock
- Too late = loss of power, overheating

Distributor vs Distributorless Systems:

- **Distributor (older):**
  - Single coil
  - Mechanical/vacuum timing advance
  - Rotating cap distributes spark to cylinders
- **Distributorless (modern):**
  - Multiple coils (1 or 2 per cylinder)
  - Electronic timing control
  - More accurate, no mechanical wear

Hour 3 (11:00 AM - 12:00 PM): Timing & Synchronization

Valve Timing Fundamentals:

- Valves must open/close at correct time relative to piston
- Camshaft controls valve timing
- Camshaft driven by crankshaft via belt or chain
- Timing marks align crank and cam positions

Why Timing Matters:

- Intake valve opens: near end of exhaust stroke (overlap)
- Intake valve closes: after BDC on compression (late closing)
- Exhaust valve opens: before BDC on power (early opening)
- Exhaust valve closes: after TDC on intake (overlap)
- Valve overlap improves breathing at certain RPM

Timing Belt/Chain:

- **Timing Belt:**
  - Rubber reinforced belt
  - Requires replacement (60,000-100,000 km)
  - Quieter operation
  - Failure = engine damage (interference engines)
- **Timing Chain:**
  - Metal chain like bicycle chain
  - Lasts longer, rarely needs replacement
  - Noisier than belt
  - Requires oil lubrication

Interference vs Non-Interference Engines:

- **Interference:**
  - Piston and valves occupy same space (at different times)
  - Belt/chain failure = piston hits valves = bent valves
  - Most modern engines are interference
- **Non-Interference:**
  - Piston never contacts valves even if timing off
  - Belt/chain failure = engine stops but no damage
  - Older designs

Timing Marks:

- Located on crankshaft pulley and camshaft sprocket

- Must align during timing belt installation
- Service manual shows exact alignment
- Use TDC marking on cylinder #1

Basic Synchronization Concept:

- Engine must coordinate:
  - Piston position (from crankshaft)
  - Valve position (from camshaft)
  - Fuel injection timing (from ECU)
  - Spark timing (from ECU)
- All based on crankshaft position sensor signal

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Fuel & Ignition Component Identification

Activity 1: Fuel System Tracing (30 min)

- Start at fuel filler cap
- Trace fuel path on actual vehicle:
  1. Fuel tank location (under vehicle)
  2. Fuel pump location (in tank - can't see)
  3. Fuel filter location (under vehicle or in engine bay)
  4. Fuel lines to engine bay
  5. Fuel rail on engine (visible)
  6. Individual injectors at each cylinder
- Draw simple fuel flow diagram
- Identify fuel pressure regulator if visible

Activity 2: Ignition System Component Identification (30 min)

- Locate ignition components:
  - **Battery** (already know this)
  - **Ignition coil(s):**
    - Modern: on top of each spark plug (coil-on-plug)
    - Older: single coil on side of engine
  - **Spark plug wires** (if present)
  - **Spark plugs** (under coils or wire boots)
  - **Crankshaft position sensor** (near crank pulley or flywheel)
- Count how many coils (= number of cylinders usually)

- Understand why modern engines have multiple coils

Pass Criteria:

- ✅ Can trace fuel path from tank to injectors
- ✅ Can locate fuel filter on vehicle
- ✅ Can identify ignition coils
- ✅ Can locate spark plugs

Hour 2 (2:00 PM - 3:00 PM): Timing Components & Fastener Identification

Activity 1: Timing Belt/Chain Observation (25 min)

- Identify if vehicle has timing belt or chain
- **Belt system:** Usually behind plastic cover
  - Locate timing belt cover
  - Understand why cover is important (protection)
  - Do NOT remove cover (not necessary today)
  - Identify belt inspection points if accessible
- **Chain system:** Inside engine (not visible)
  - Identify chain cover (usually metal)
  - Listen to engine running: chain makes light noise
- Locate timing marks on crankshaft pulley
- Understand these marks align with reference

Activity 2: Fastener Type Identification (35 min)

- Gather examples of different fastener types
- **Bolts vs Screws:**
  - Bolt: goes through hole, needs nut
  - Screw: threads into tapped hole
- **Head types:**
  - Hex head (most common)
  - Phillips head
  - Torx head (T20, T25, T30, etc.)
  - Allen/Hex socket
  - 12-point (double hex)
- **Thread types:**
  - Coarse thread (standard)
  - Fine thread (more threads per mm)
- **Special fasteners:**

  - Flange bolt (built-in washer)
  - Cap screw (high strength)
  - Stud bolt (threaded both ends)

Bolt Grade Identification:

- **Metric bolt grades:**
  - Marked on head: 8.8, 10.9, 12.9
  - Higher number = stronger
  - 8.8 most common in automotive
- **Measuring bolts:**
  - Diameter: across threads (e.g., M8 = 8mm)
  - Length: under head to end
  - Pitch: distance between threads (e.g., 1.25mm)

Pass Criteria:

- ✅ Can identify timing belt vs chain system
- ✅ Can locate timing marks on crankshaft pulley
- ✅ Can identify different fastener head types
- ✅ Can read bolt grade markings

Hour 3 (3:00 PM - 4:00 PM): Tightening Sequence Practice

Activity 1: Understanding Torque Sequence (20 min)

- Why sequence matters:
  - Prevents warping of mating surfaces
  - Distributes clamping force evenly
  - Critical for head gaskets, intake manifolds

Common Sequences:

```
Cylinder head pattern (4-bolt):
    2---4
    |   |
    1---3
Sequence: 1-3-4-2

Cylinder head pattern (6-bolt):
    3---5
    |   |
```

```
    1---2---6---4
Sequence: 1-2-3-4-5-6 (inside to outside)
```

- Always start from center, work outward
- OR in cross pattern (star pattern)
- Never tighten one side completely first

Activity 2: Practice on Valve Cover (20 min)

- Use valve cover or practice fixture
- Mark bolt positions
- Practice tightening sequence:
  1. Install all bolts finger-tight
  2. Identify starting position (center or pattern)
  3. Tighten 1/4 turn in sequence
  4. Repeat sequence, another 1/4 turn
  5. Repeat until properly tight
- Understand multi-pass tightening

Activity 3: Introduction to Torque Wrench (20 min)

- **Types of torque wrenches:**
  - Click-type (most common)
  - Beam-type (mechanical)
  - Digital (expensive)

Click-Type Operation:

- Set desired torque value on handle
- Install socket
- Tighten until you hear/feel "click"
- Stop immediately when it clicks
- Don't continue pulling after click

Basic Practice:

- Set torque wrench to 20 Nm
- Tighten practice bolts to feel the click
- Understand "hand tight" vs "torque spec"
- Read torque specification from service manual

Torque Specifications:

- Where to find: service manual, bolt databases

- Typically in Nm (Newton-meters) or lb-ft
- Example: M8 bolt = 20-25 Nm
- Example: M10 bolt = 40-50 Nm
- Always use correct specification

Pass Criteria:

- ✅ Can explain why sequence matters
- ✅ Can demonstrate cross/star pattern
- ✅ Can set torque wrench to specified value
- ✅ Can tighten bolt until click

DAY 3 ASSESSMENT:

Practical Skills Checklist:

- [ ] Traces complete fuel system path
- [ ] Identifies ignition components
- [ ] Identifies timing belt/chain system
- [ ] Recognizes different fastener types
- [ ] Demonstrates proper tightening sequence
- [ ] Uses torque wrench at basic level

Knowledge Check (Verbal):

- Name 4 components in fuel system path
- What does the ignition coil do?
- Why must timing belt be replaced?
- Why do we tighten bolts in a sequence?

DAY 4: INTAKE SYSTEM + BASIC MEASUREMENTS

Learning Objectives:

- Understand airflow principles and intake path
- Identify intake system components
- Use basic measuring tools (tape measure, calipers)
- Understand metric measurements and specifications

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Airflow Principles

Why Engine Needs Air:

- Combustion requires oxygen (21% of air)
- Fuel needs 14.7 parts air to burn completely
- More air = more fuel = more power
- Engine is essentially an air pump

Intake System Purpose:

- Provide clean filtered air
- Measure air quantity (MAF/MAP sensor)
- Control airflow (throttle body)
- Distribute evenly to all cylinders

Airflow Path (Complete Journey):

1. **Air filter housing:**
   - Outside air enters
   - Large opening for maximum flow
   - Pre-filter in some vehicles
2. **Air filter element:**
   - Paper or foam element
   - Traps dust, dirt, debris
   - Must flow freely when clean
3. **Air intake hose:**
   - Connects filter to throttle body
   - Often has resonator (noise reduction)
4. **Mass Airflow (MAF) sensor OR MAP sensor:**
   - Measures air quantity
   - Signals ECU for fuel calculation
5. **Throttle body:**
   - Butterfly valve controls airflow
   - Opened by accelerator pedal
   - Idle air control valve or electronic control
6. **Intake manifold:**
   - Distributes air to cylinders
   - Plenum (large chamber) evens out pulses
   - Runners to each cylinder

7. **Intake port in cylinder head:**
    - Final path to cylinder
8. **Intake valve:**
    - Opens to let air in

Volumetric Efficiency:

- Engine's ability to fill cylinders with air
- Naturally aspirated: ~80-90% at best
- Limited by intake restrictions
- Why forced induction helps (coming Day 6)

Hour 2 (10:00 AM - 11:00 AM): Intake Components & Function

Air Filter Types:

- **Paper element (most common):**
    - Pleated paper, high surface area
    - Replace per service schedule (10,000-15,000 km)
    - Cannot be cleaned effectively
- **Foam/Oil-bath:**
    - Older vehicles, some motorcycles
    - Can be cleaned and reused
- **High-flow performance filters:**
    - Cotton gauze, oiled
    - Reusable but need re-oiling

Air Filter Inspection:

- Check for dirt accumulation
- Look for tears or damage
- Ensure proper seating in housing
- Dirty filter = reduced power, fuel economy

MAF (Mass Airflow) Sensor:

- Hot wire or hot film type
- Measures actual air mass (not volume)
- Critical for fuel calculation
- Very sensitive, can fail if contaminated
- Located between filter and throttle body

MAP (Manifold Absolute Pressure) Sensor:

- Alternative to MAF (not both)
- Measures vacuum/pressure in intake manifold
- ECU calculates airflow from pressure
- Less sensitive than MAF, more reliable

Throttle Body:

- Houses butterfly valve
- **Cable-operated (older):**
  - Physical cable from accelerator pedal
  - Mechanical connection
- **Electronic (Drive-by-Wire):**
  - Pedal position sensor only
  - ECU controls throttle motor
  - More common now
- Idle air control:
  - Bypass passage around throttle
  - Controls idle speed

Intake Manifold:

- **Design considerations:**
  - Long runners = low RPM torque
  - Short runners = high RPM power
  - Variable length systems exist
- **Materials:**
  - Aluminum (traditional)
  - Plastic composite (modern, lighter)
- **Features:**
  - Vacuum ports for brake booster, etc.
  - PCV connection (crankcase ventilation)
  - EGR passage (emission control)

Intake Air Temperature (IAT) Sensor:

- Measures incoming air temperature
- Cold air = more dense = more power
- Usually integrated in MAF sensor
- Sometimes separate in intake manifold

Hour 3 (11:00 AM - 12:00 PM): Intake Problems & Symptoms

Common Intake System Issues:

1. Air Filter Problems:

- **Dirty filter:**
  - Reduced airflow
  - Loss of power
  - Poor fuel economy
  - Black smoke (rich mixture)
- **Missing or damaged filter:**
  - Dirt enters engine
  - MAF sensor contamination
  - Accelerated engine wear

2. Air Leaks (Unmetered Air):

- Occurs after MAF sensor
- ECU doesn't know about this air
- Causes lean mixture
- **Symptoms:**
  - Rough idle
  - Check engine light (lean codes)
  - Hesitation on acceleration
- **Common leak locations:**
  - Intake hose cracks
  - Throttle body gasket
  - Manifold gasket
  - Vacuum hose cracks

3. Throttle Body Issues:

- Carbon buildup on throttle plate
- Sticking throttle valve
- Electronic throttle motor failure
- **Symptoms:**
  - Rough idle
  - Stalling
  - Poor throttle response

4. MAF Sensor Problems:

- Contamination (oil, dirt)
- Failure of hot wire
- **Symptoms:**
  - Poor fuel economy
  - Black smoke
  - Hesitation
  - Check engine light

5. Intake Manifold Gasket Leak:

- Vacuum leak at cylinder head interface
- **Symptoms:**
  - Rough idle (worse when cold)
  - Coolant leak (if water-heated manifold)
  - Whistling sound

Visual Inspection Checklist:

- Air filter condition
- Intake hose cracks or loose clamps
- Vacuum hoses for cracks
- Throttle body cleanliness
- Manifold gasket oil/coolant leaks
- Listen for whistling (air leak)

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Intake System Mapping

Activity 1: Complete Air Path Tracing (30 min)

- Start from air filter housing intake snorkel
- Trace complete path on actual vehicle:
  1. Outside air entry point
  2. Air filter housing
  3. Air filter element (remove to inspect)
  4. MAF sensor location (if present)
  5. Intake hose to throttle body
  6. Throttle body position
  7. Intake manifold
  8. Individual runners to cylinders

- Take photos at each point
- Draw airflow diagram with all components

Activity 2: Component Identification & Labeling (30 min)

- Create detailed map of intake system
- Label each component:
  - Air filter housing
  - Air filter element
  - MAF or MAP sensor
  - Intake air temperature sensor
  - Throttle body
  - Idle air control valve (if present)
  - Intake manifold
  - PCV valve location
  - Vacuum hoses and connections
- Note any cracks, damage, or leaks observed

Pass Criteria:

- ✅ Can trace complete intake air path
- ✅ Can identify all major intake components
- ✅ Can locate MAF/MAP sensor
- ✅ Creates accurate system diagram

Hour 2 (2:00 PM - 3:00 PM): Basic Measurement Skills

Activity 1: Tape Measure Practice (20 min)

- Using steel tape measure
- **Metric system basics:**
  - 1 meter = 100 centimeters
  - 1 centimeter = 10 millimeters
  - Reading mm graduations
- Practice measuring:
  - Engine bay dimensions
  - Component lengths
  - Hose diameters
  - Belt lengths
- Record measurements accurately
- Understand tolerance (± variation)

Activity 2: Digital Caliper Introduction (20 min)

- **Caliper components:**
  - Main scale
  - Vernier scale (or digital display)
  - Outside jaws (external dimensions)
  - Inside jaws (internal dimensions)
  - Depth probe
  - Locking screw

Using Digital Calipers:

- Zero the caliper before measuring
- Close jaws gently on object
- Read display (typically to 0.01mm)
- Lock for recording measurement
- Open and zero for next measurement

Practice Measurements:

- Bolt diameter measurement
- Nut size measurement
- Hose inside diameter
- Small component dimensions
- Understanding precision vs accuracy

Activity 3: Specification Understanding (20 min)

- Reading service manual specifications
- Example: "Intake valve clearance: 0.15-0.25mm"
  - Minimum: 0.15mm
  - Maximum: 0.25mm
  - Middle of range preferred: 0.20mm
- Example: "Air filter replacement: 10,000 km ± 1,000 km"
  - Acceptable range: 9,000-11,000 km
- Understanding "within specification" vs "out of spec"

Pass Criteria:

- ✅ Can read tape measure to nearest mm
- ✅ Can operate digital calipers
- ✅ Measures objects with reasonable accuracy
- ✅ Records measurements properly

Hour 3 (3:00 PM - 4:00 PM): Intake System Inspection Practice

Activity 1: Air Filter Inspection & Replacement (20 min)

- Remove air filter housing cover
  - Identify clips or screws
  - Use correct tools
  - Support housing when opening
- Remove air filter element
- Inspect condition:
  - Check pleats for dirt loading
  - Look for tears or holes
  - Check sealing edges
  - Hold up to light to check restriction
- Determine: Replace or acceptable?
- Install filter (correct orientation)
- Secure housing properly

Activity 2: Intake Hose & Clamp Inspection (20 min)

- Inspect all intake hoses:
  - Look for cracks (bend hoses gently)
  - Check for collapsed sections
  - Look for oil contamination
  - Feel for soft spots (deterioration)
- Check hose clamps:
  - Screw-type clamps: tight?
  - Spring clamps: holding properly?
  - Look for missing clamps
- Check MAF sensor wiring:
  - Connector secure?
  - Wires damaged?

Activity 3: Basic Restriction Check (20 min)

- **Understanding restrictions:**
  - Dirty filter restricts airflow
  - Collapsed hose restricts airflow
  - Can measure with manometer (pressure gauge)
  - Basic check: engine running, remove air filter
    - Should see MAF sensor or throttle body

    - No visible restrictions?

Throttle Body Observation:

- Remove intake hose from throttle body
- Observe throttle plate:
  - Clean or carbon buildup?
  - Moves freely by hand? (key off!)
  - Returns to closed position?
- **DO NOT CLEAN TODAY** (requires specific procedure)
- Just observe and note condition

Document Findings:

- Create inspection report:
  - Air filter condition (replace/OK)
  - Hose condition (all locations)
  - Clamp security
  - Throttle body condition
  - Any leaks observed
- Understand this is basic preventive maintenance

Pass Criteria:

- ✅ Can remove and inspect air filter
- ✅ Can identify hose cracks/damage
- ✅ Can check clamp tightness
- ✅ Creates basic inspection report

DAY 4 ASSESSMENT:

Practical Skills Checklist:

- [ ] Traces complete intake air path
- [ ] Identifies all intake system components
- [ ] Uses tape measure accurately
- [ ] Operates digital calipers correctly
- [ ] Inspects air filter condition
- [ ] Checks intake hoses for damage
- [ ] Documents findings in report format

Knowledge Check (Verbal):

- What does the MAF sensor do?
- Why must air be filtered?
- What happens if there's an air leak after the MAF sensor?
- What is volumetric efficiency?

DAY 5: EXHAUST SYSTEM + COMPONENT REMOVAL PRACTICE

Learning Objectives:

- Understand exhaust flow and back pressure
- Identify emission control components
- Master component removal techniques
- Understand exhaust safety considerations

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Exhaust Flow Theory

Exhaust System Purpose:

- Remove burned gases from cylinders
- Reduce exhaust noise
- Control emissions (pollution)
- Maintain optimal back pressure
- Route hot gases safely away from vehicle

Exhaust Flow Path:

1. **Exhaust Port (cylinder head):**
   - Exhaust valve opens
   - High-temperature gases (500-900°C) exit
2. **Exhaust Manifold:**
   - Collects gases from all cylinders
   - Cast iron or stainless steel
   - Multiple runners merge into single outlet
3. **Oxygen Sensor (before catalyst):**
   - Measures oxygen in exhaust
   - Helps ECU adjust fuel mixture

4. **Catalytic Converter:**
   - Chemical reaction cleans exhaust
   - Reduces CO, HC, NOx emissions
   - Contains precious metals (platinum, palladium)
   - Gets very hot (400-800°C)
5. **Oxygen Sensor (after catalyst):**
   - Monitors catalyst efficiency
6. **Resonator (if present):**
   - Secondary muffler
   - Fine-tunes exhaust note
7. **Main Muffler/Silencer:**
   - Primary noise reduction
   - Expansion chambers and baffles
   - Absorptive packing in some designs
8. **Tailpipe:**
   - Exits under/behind vehicle
   - Directs gases away from occupants

Back Pressure Concept:

- Some resistance to flow is needed
- Too much = power loss, poor efficiency
- Too little = power loss (no scavenging effect)
- Optimal back pressure helps:
  - Pull exhaust out of cylinder (scavenging)
  - Improve low-end torque
  - Reduce valve overlap issues

Exhaust Gas Characteristics:

- Very hot (500-900°C at manifold)
- Contains:
  - Nitrogen (N_2) - 70%
  - Carbon dioxide (CO_2) - 12-15%
  - Water vapor (H_2O) - 10-12%
  - Carbon monoxide (CO) - small amount
  - Hydrocarbons (HC) - small amount
  - Nitrogen oxides (NOx) - small amount
  - Particulate matter (PM) - especially diesel
- Toxic: never run engine in closed space!

Hour 2 (10:00 AM - 11:00 AM): Exhaust Components Deep Dive

Exhaust Manifold:

- **Cast iron (traditional):**
  - Heavy, durable
  - Good heat retention
  - Can crack from thermal cycling
- **Stainless steel tubular headers:**
  - Lighter weight
  - Better flow (performance)
  - Expensive
  - Less common on stock vehicles
- **Design factors:**
  - Equal length runners (performance)
  - Collector design
  - Thermal expansion considerations
- **Common problems:**
  - Cracks from heat cycles
  - Warping
  - Gasket leaks (ticking sound)
  - Broken studs

Catalytic Converter:

- **How it works:**
  - Ceramic honeycomb substrate
  - Coated with precious metals
  - Chemical reactions at high temperature:
    - $CO + O_2 \rightarrow CO_2$ (oxidation)
    - $HC + O_2 \rightarrow CO_2 + H_2O$ (oxidation)
    - $NO_x \rightarrow N_2 + O_2$ (reduction)
  - Requires 300°C minimum (light-off temperature)
- **Types:**
  - **Three-way catalyst (TWC):** Most common, petrol engines
  - **Diesel oxidation catalyst (DOC):** Diesel engines
  - **Diesel particulate filter (DPF):** Traps soot
- **Location:**
  - Close-coupled: near engine (faster warm-up)
  - Underfloor: further from engine
  - Some vehicles have multiple converters

- **Common issues:**
  - Contamination from oil/coolant
  - Physical damage from road debris
  - Theft (precious metals valuable)
  - Clogging (especially DPF)

Oxygen Sensors (O2/Lambda Sensors):

- **Before catalyst (upstream):**
  - Primary sensor for fuel control
  - Switches between rich and lean
  - Helps maintain stoichiometric ratio (14.7:1)
- **After catalyst (downstream):**
  - Monitors catalyst efficiency
  - Should show steady signal (catalyst working)
  - Varying signal = catalyst failing
- **Types:**
  - Zirconia (switching type): older
  - Wideband (linear): modern, more precise
- **Wiring:**
  - Heated (3-4 wires): most common
  - Unheated (1-2 wires): older

Muffler/Silencer:

- **Noise reduction methods:**
  - **Reflection:** Bounces sound waves
  - **Absorption:** Fiberglass/steel wool packing
  - **Expansion:** Large chambers slow gases
- **Types:**
  - **Reactive (chambered):** Baffles and chambers
  - **Absorptive (turbo-style):** Straight-through with packing
  - **Combination:** Most common
- **Common problems:**
  - Rust perforation (water condensation)
  - Internal baffle collapse
  - Packing deterioration (gets louder)

Exhaust Hangers & Mounts:

- Rubber isolators absorb vibration
- Allow thermal expansion

- Common failure: rubber deteriorates, exhaust rattles
- Exhaust should not be solid-mounted

Hour 3 (11:00 AM - 12:00 PM): Emission Standards & Safety

Indian Emission Standards:

- **BS4 (Bharat Stage 4):**
  - Implemented 2017-2020
  - Required catalytic converter
  - On-board diagnostics (OBD-II)
- **BS6 (Bharat Stage 6):**
  - Implemented April 2020
  - Much stricter limits
  - Requires:
    - Advanced catalytic converters
    - Particulate filters (diesel)
    - More sensors
    - Better fuel quality

Emission Limits (simplified):

- CO (Carbon Monoxide): < 1.0 g/km
- HC (Hydrocarbons): < 0.1 g/km
- NOx (Nitrogen Oxides): < 0.06 g/km (petrol), < 0.08 g/km (diesel)
- PM (Particulate Matter): < 0.0045 g/km (diesel)

Exhaust Safety Considerations:

1. Carbon Monoxide (CO) Danger:

- Colorless, odorless gas
- Deadly in enclosed spaces
- Binds to hemoglobin (prevents oxygen transport)
- **Never run engine in closed garage!**
- Symptoms of exposure:
  - Headache, dizziness
  - Nausea
  - Unconsciousness
  - Death

2. Heat Hazards:

- Exhaust components extremely hot
- **Manifold:** 500-900°C
- **Catalytic converter:** 400-800°C
- **Muffler:** 200-400°C
- Severe burn risk
- Can ignite oil, fuel, flammable materials
- **Never touch exhaust after engine running**
- Allow proper cool-down time

3. Toxic Fumes:

- Besides CO, exhaust contains:
  - NOx: respiratory irritant
  - HC: carcinogenic
  - PM: respiratory damage
- Always work in ventilated area
- Use exhaust extraction system when possible

4. Working Under Vehicle:

- Exhaust work often requires going underneath
- Ensure proper lifting and support
- Let exhaust cool completely first
- Wear protective gloves
- Have fire extinguisher nearby

Visual Inspection Indicators:

- **Rust/corrosion:** Component age, water condensation
- **Black soot:** Rich mixture, oil burning
- **White deposits:** Coolant entering exhaust (head gasket)
- **Broken hangers:** Exhaust rattling, misalignment
- **Physical damage:** Road debris, scraping
- **Leaks:** Listen for hissing, check connections

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Exhaust System Mapping & Identification

Activity 1: Exhaust Path Tracing (30 min)

- **SAFETY FIRST:** Ensure vehicle is cold!
- Lift vehicle safely on hoist/stands
- Trace complete exhaust path:
  1. Exhaust manifold (engine side)
  2. Down pipe/front pipe
  3. Oxygen sensor #1 (before catalyst)
  4. Catalytic converter
  5. Oxygen sensor #2 (after catalyst)
  6. Center pipe/intermediate pipe
  7. Resonator (if present)
  8. Muffler/silencer
  9. Tailpipe
- Take photos from underneath
- Draw exhaust system diagram

Activity 2: Component Identification (30 min)

- Locate and identify:
  - **Exhaust manifold:** Count how many runners
  - **Gasket:** Between manifold and head
  - **Studs/bolts:** Note any broken or rusted
  - **Flex pipe:** Allows movement (if present)
  - **Catalytic converter:** Large bulge in pipe
    - Note if it's close-coupled or underfloor
    - Identify heat shields
  - **Oxygen sensors:** Count how many (usually 2)
    - Note wiring harness routing
  - **Muffler:** Main silencer box
  - **Hangers:** Rubber mounts holding exhaust
- Check for:
  - Rust/corrosion
  - Physical damage (dents)
  - Broken hangers
  - Soot deposits (leak indicators)

Pass Criteria:

- ✅ Can trace complete exhaust path
- ✅ Can identify catalytic converter
- ✅ Can locate oxygen sensors
- ✅ Can identify exhaust hangers

Hour 2 (2:00 PM - 3:00 PM): Systematic Component Removal Practice

Activity 1: Battery Removal & Installation (25 min)

- **Safety briefing:**
  - Which terminal first? NEGATIVE (black) first on removal
  - Which terminal first on installation? POSITIVE (red) first
  - Why? Prevents short circuit through tools
  - Never touch both terminals simultaneously
  - Remove jewelry (rings conduct electricity)

Removal Procedure:

1. Turn off all electrical loads
2. Turn off ignition, remove key
3. Locate battery terminals
4. Identify terminal sizes (usually 10mm)
5. Loosen NEGATIVE terminal first
6. Remove negative cable, secure away from terminal
7. Loosen POSITIVE terminal
8. Remove positive cable
9. Remove hold-down bracket (already practiced Day 1)
10. Lift battery carefully (heavy! ~12-15 kg)
11. Inspect terminals for corrosion

Installation Procedure (REVERSE ORDER):

1. Place battery in tray
2. Install hold-down bracket
3. Clean terminals if corroded (wire brush)
4. Connect POSITIVE terminal first
5. Tighten positive connection
6. Connect NEGATIVE terminal
7. Tighten negative connection
8. Apply terminal protectant (grease/spray)
9. Check: terminals tight, nothing loose

Activity 2: Engine Cover Removal (20 min)

- Many modern engines have plastic covers
- **Purpose of cover:**
  - Sound dampening

  - Aesthetic appearance
  - Dust/dirt protection
  - Minor thermal insulation

Removal types:

- **Clip-type:** Pull straight up
- **Bolt-type:** Remove bolts first
- **Combination:** Bolts + clips
- Identify mounting method first
- Use correct technique for type

Practice removal:

- Identify all attachment points
- Remove bolts if present
- Pull clips gently but firmly
- Don't force (can break clips)
- Store clips with cover
- Reinstall practice: align clips first

Activity 3: Fluid Cap Removal & Inspection (15 min)

- **Oil filler cap:**
  - Turn counter-clockwise
  - Check gasket condition
  - Look inside: sludge? Normal color?
- **Coolant reservoir cap:**
  - **ONLY when cold!** Never hot!
  - Note pressure rating on cap
  - Inspect for cracks
- **Brake fluid reservoir cap:**
  - Wipe clean before opening
  - Check fluid level
  - Note MIN/MAX marks
- **Power steering reservoir (if present):**
  - Check level
  - Note dipstick if equipped

Pass Criteria:

- ✅ Can remove battery safely (correct sequence)
- ✅ Can install battery safely (correct sequence)

- ✅ Can remove engine cover without damage
- ✅ Can check fluid levels safely

Hour 3 (3:00 PM - 4:00 PM): Organized Disassembly Technique

Activity 1: Parts Organization System (20 min)

- **Why organization matters:**
  - Prevents lost parts
  - Speeds reassembly
  - Looks professional
  - Prevents mistakes

Organization methods:

- **Small containers/trays:** Group related parts
- **Magnetic parts tray:** Holds fasteners
- **Labeled bags:** Write location on bag
- **Parts tray with dividers:** One area per component
- **Photo documentation:** Before/during disassembly

Practice organizing:

- Given set of mixed fasteners and parts
- Sort by:
  - Size
  - Type
  - Location they came from
- Label containers
- Create simple inventory list

Activity 2: Fastener Tracking Method (20 min)

- **Problem:** Mixing up bolts during disassembly
- **Solution:** Systematic tracking

Techniques:

- **Tape method:** Use masking tape, stick bolts to tape, label
- **Diagram method:** Draw component, place bolts on diagram
- **Container method:** One container per assembly
- **Threading back method:** Thread bolts back into holes

Practice:

- Remove multiple bolts from different locations
- Use one tracking method
- Store systematically
- Reinstall from stored location
- Verify all correct sizes/locations

Activity 3: Photo Documentation Practice (20 min)

- **Why photos help:**
  - Remember wire routing
  - Remember hose connections
  - Remember bracket positions
  - Reference for reassembly
  - Evidence of prior condition

Good documentation photos:

- Before disconnecting anything
- Overall view showing everything
- Close-up of connections
- Multiple angles
- Include reference (hand for scale)
- Good lighting

Practice documentation:

- Photograph engine bay before any work
- Photograph specific area (intake system)
- Take detail photos of connections
- Review photos: Are they useful?
- Discuss what makes good vs bad photos

Real-World Application:

- Instructor shows complex repair photos
- Discuss how photos saved time
- Examples of repairs that went wrong (no photos)
- Professional habit: always document

Pass Criteria:

- ✅ Can organize removed parts systematically

- ✅ Can track fasteners to prevent mixing
- ✅ Takes useful documentation photos
- ✅ Understands reverse-sequence reassembly

DAY 5 ASSESSMENT:

Practical Skills Checklist:

- [ ] Traces complete exhaust system path
- [ ] Identifies emission control components
- [ ] Removes battery with correct sequence
- [ ] Removes engine cover without damage
- [ ] Organizes removed parts systematically
- [ ] Tracks fasteners to prevent mixing
- [ ] Documents work with photos

Knowledge Check (Verbal):

- What does the catalytic converter do?
- Why is carbon monoxide dangerous?
- Which battery terminal do you disconnect first, and why?
- Why must we organize parts during disassembly?

DAY 6: FORCED INDUCTION AWARENESS + TOOL ORGANIZATION

Learning Objectives:

- Understand turbocharger and supercharger concepts (awareness level)
- Identify forced induction components on vehicles
- Master professional tool organization
- Understand tool maintenance and care

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Forced Induction Fundamentals

Why Forced Induction?

- Natural aspiration limited by atmospheric pressure
- Can only fill cylinders to ~80-90% capacity
- Forced induction pushes more air in
- More air = more fuel = more power
- Get big engine power from small engine

Two Types:

1. TURBOCHARGER (Exhaust-Driven):

- Uses exhaust gas energy (waste energy recovery)
- Turbine wheel spins in exhaust
- Compressor wheel on same shaft compresses intake air
- Spins at 100,000-250,000 RPM!
- "Free" power (uses waste energy)

Components:

- Turbine housing (exhaust side)
- Turbine wheel (spins from exhaust)
- Center bearing housing (oil-cooled bearings)
- Compressor housing (intake side)
- Compressor wheel (compresses air)
- Wastegate (controls boost pressure)

2. SUPERCHARGER (Belt-Driven):

- Mechanically driven by engine belt
- Direct connection to crankshaft
- No lag (instant boost)
- Uses engine power to make more power (parasitic loss)
- Less common than turbo

Types of superchargers:

- Roots-type (older, less efficient)
- Twin-screw (better efficiency)
- Centrifugal (looks like turbo compressor)

Hour 2 (10:00 AM - 11:00 AM): Forced Induction Systems

Boost Pressure Concepts:

- Measured in:
  - PSI (pounds per square inch)
  - Bar (1 bar = 14.7 PSI = atmospheric)
  - kPa (kilopascals)
- Typical boost: 0.5-1.5 bar (7-22 PSI)
- High performance: 2+ bar
- Pressure measured relative to atmosphere (gauge pressure)

Supporting Components:

Intercooler:

- **Why needed:** Compression heats air
- Hot air less dense (defeats purpose)
- Intercooler cools compressed air
- **Types:**
  - Air-to-air: Airflow through radiator-like core
  - Air-to-water: Water cooling system (less common)
- Mounted in front of radiator or on top of engine

Wastegate:

- Controls maximum boost pressure
- Bleeds exhaust around turbine at target pressure
- **Internal:** Built into turbo housing
- **External:** Separate valve in exhaust
- Prevents over-boost (engine damage)

Blow-Off Valve (BOV) / Bypass Valve:

- Releases pressure when throttle closes
- Prevents compressor surge
- Protects turbo bearings
- **BOV:** Vents to atmosphere (whoosh sound)
- **Bypass valve:** Recirculates to intake (quieter, emissions-friendly)

Turbocharger Lubrication:

- Requires engine oil supply
- Oil feed line from engine to turbo
- Oil return line from turbo to oil pan
- Critical: Oil must be clean and flow freely
- Turbo failure often from oil problems

Turbocharger Cooling:

- Most turbos oil-cooled only
- Some have water cooling jacket
- Gets extremely hot (exhaust-side)
- Must idle briefly after hard driving (cool-down)

Hour 3 (11:00 AM - 12:00 PM): Turbo Systems & Awareness

Turbo Lag:

- Delay between pressing throttle and boost building
- Time for turbo to spin up
- Depends on:
  - Turbo size (larger = more lag)
  - Engine RPM (higher = less lag)
  - Exhaust energy
- Modern turbos reduce lag with technology:
  - Variable geometry turbos (VGT)
  - Twin-scroll designs
  - Better bearings

Variable Geometry Turbo (VGT):

- Adjustable vanes in turbine housing
- Small opening = fast spool at low RPM
- Large opening = high flow at high RPM
- Computer-controlled
- Common on diesel engines

Twin-Turbo Systems:

- **Sequential:** Small turbo for low RPM, large for high RPM
- **Parallel:** Two identical turbos, one per bank (V-engines)
- Reduces lag, increases power range

Common Issues (Awareness Level):

- **Oil starvation:** Bearing failure, turbo death
- **Foreign object damage:** Something sucked into compressor
- **Boost leaks:** Hose connections, intercooler damage
- **Wastegate failure:** Over-boost or under-boost

- **Shaft play:** Bearings worn, noise, oil in intake
- **Compressor surge:** Fluttering sound when throttle closed

Turbocharged Engine Considerations:

- Must use specified oil (often synthetic)
- Oil changes more critical
- Premium fuel often required (higher octane)
- Let engine idle before shutdown (cool-down)
- Intercooler maintenance (cleanliness)
- More complex than N/A engine
- Higher maintenance cost but more power/efficiency

Forced Induction Identification on Vehicle:

- Look for intercooler (front of vehicle or top of engine)
- Larger intake/exhaust plumbing
- "Turbo" or "T" badge
- Boost gauge on dashboard
- Higher fuel requirements (sticker)
- Hear turbo sound (whoosh, whistle)

Why This is "Awareness Level":

- We're not rebuilding turbos in Fundamentals
- Need to know it exists and basic function
- Can identify on vehicle
- Deep dive = Mechanical Specialization

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Forced Induction Component Identification

Activity 1: Turbocharged Vehicle Inspection (If Available) (30 min)

- Locate turbocharged vehicle
- Identify components:
  - **Turbocharger location:** Usually attached to exhaust manifold
  - **Intercooler:** Front-mounted or top-mounted
  - **Boost piping:** Connections from turbo to intercooler to intake
  - **Oil feed line:** Small line from engine to turbo
  - **Oil return line:** Larger line from turbo to oil pan

    - **Wastegate actuator:** Vacuum/pressure capsule
    - **Blow-off valve:** Listen for location
- Note differences from naturally aspirated engine
- Take photos, create component map

Activity 2: Non-Turbocharged Vehicle Comparison (30 min)

- Compare same area on N/A vehicle
- Notice:
    - Simpler intake plumbing
    - No intercooler
    - No turbo (obviously)
    - Different engine sound
- Understand complexity difference
- Discuss why turbo adds cost but also efficiency

If No Turbo Vehicle Available:

- Watch video demonstration
- Study photos of turbo installations
- Examine turbocharger cut-away model
- Identify components on display turbo

Pass Criteria:

- ✅ Can identify turbo vs non-turbo engine
- ✅ Can locate intercooler (if present)
- ✅ Understands basic boost flow path
- ✅ Can explain turbocharger purpose at basic level

Hour 2 (2:00 PM - 3:00 PM): Professional Tool Organization

Activity 1: Tool Box Organization Principles (20 min)

- **Why organization matters:**
    - Saves time (know where everything is)
    - Prevents loss (expensive tools protected)
    - Professional appearance
    - Safety (tools not scattered)
    - Productivity (estimated 30% time saving)

Organization strategies:

- **By size:** Smallest to largest
- **By frequency:** Most-used tools most accessible
- **By function:** All sockets together, all spanners together
- **Shadow board:** Outline of each tool's position
- **Color coding:** Different colors for different sizes/types

Demonstration:

- Show well-organized tool box
- Show poorly organized tool box
- Discuss which is more professional
- Time test: Find specific tool in each

Activity 2: Creating Personal Tool Organization (25 min)

- Each student receives tool set
- Create organization plan:
  - Decide drawer/section assignments
  - Group related tools
  - Place frequently-used tools in easy reach
  - Deep drawers for heavy tools
  - Shallow drawers for small items
- Implement plan
- Practice retrieving tools by name
- Instructor calls out tool, student retrieves it

Tool Inventory Management:

- Create tool inventory list
- Check tools in/out at day start/end
- Missing tool = find it or replace it
- Never leave tools in vehicle (theft risk)
- Company tools vs personal tools

Activity 3: Proper Tool Storage (15 min)

- **Hand tools:**
  - Clean before storing
  - Light oil coating (prevents rust)
  - Store in dry place
  - Magnetic strip for wrenches
  - Socket rails for sockets

Power tools:

- Remove battery when storing
- Coil cords properly
- Store in case if provided
- Check for damage before storing

Measuring tools:

- Most delicate tools
- Separate storage (won't get dropped on)
- Return to case after use
- Keep calibration records

Pass Criteria:

- ✅ Organizes tool box systematically
- ✅ Can locate any tool within 10 seconds
- ✅ Understands tool inventory process
- ✅ Stores tools properly

Hour 3 (3:00 PM - 4:00 PM): Tool Maintenance & Care

Activity 1: Hand Tool Maintenance (20 min)

- **Cleaning:**
  - Remove oil, grease, dirt after use
  - Use parts cleaner or degreaser
  - Dry thoroughly
  - Light oil coating for storage

Inspection for wear/damage:

- **Spanners/wrenches:**
  - Open-end: check for spreading
  - Ring-end: check for rounding
  - Worn = rounds off fasteners = discard
- **Sockets:**
  - Check for cracks
  - Check for rounding of internal drive
  - Worn = replace
- **Ratchets:**

    - Test pawl operation (both directions)
    - Check for slipping
    - Lubricate with light oil if stiff
- **Screwdrivers:**
    - Check tip wear
    - Worn tip = cam-out = damaged screw
    - Regrind tip or replace

Activity 2: Tool Damage Prevention (20 min)

- **Don't:**
    - Use tools as hammers (except hammers)
    - Use cheater bars on ratchets (breaks mechanism)
    - Leave tools in rain/moisture
    - Drop tools (cracks sockets)
    - Mix dirty tools with clean ones
    - Use wrong tool for job (rounds fasteners)

Do:

- Use correct size every time
- Apply force smoothly, not sudden
- Use penetrating oil on stuck fasteners
- Keep tools clean and dry
- Inspect before each use
- Replace worn tools promptly

Activity 3: Torque Wrench Special Care (20 min)

- **Most important tool care:**
    - Torque wrench is precision instrument
    - Must be calibrated regularly (annually recommended)
    - Expensive to replace if damaged

Storage:

- ALWAYS return to lowest setting after use
    - Spring remains relaxed
    - Prevents calibration drift
- Store in case horizontally
- Never drop
- Never use as breaker bar

Calibration check:

- Should be done professionally
- Can do basic check:
  - Known weight at known distance
  - Calculate expected torque
  - Compare to wrench reading
- Out of calibration = send for service

Cleaning:

- Wipe clean after use
- Never submerge in cleaner
- Don't oil calibration adjustment
- Keep clean and dry

Using torque wrench correctly:

- Pull smoothly, don't jerk
- Pull perpendicular to wrench handle
- Stop immediately at click
- One click = correct torque
- Don't continue pulling after click
- Reset to lowest setting afterward

Pass Criteria:

- ✅ Can clean tools properly
- ✅ Can inspect tools for wear
- ✅ Understands torque wrench care
- ✅ Stores tools correctly for longevity

DAY 6 ASSESSMENT:

Practical Skills Checklist:

- [ ] Can identify turbocharged vs N/A engine
- [ ] Can locate forced induction components
- [ ] Organizes tool box professionally
- [ ] Can retrieve any tool quickly
- [ ] Cleans tools after use
- [ ] Inspects tools for damage

- [ ] Stores torque wrench correctly

Knowledge Check (Verbal):

- What is the purpose of forced induction?
- Name two types of forced induction.
- What does an intercooler do?
- Why must we return torque wrench to lowest setting?
- Why is tool organization important?

WEEK 1 SUMMARY & OUTCOMES

Week 1 Learning Journey:

"From Complete Beginner to Basic Mechanical Literacy"

Day 1: Workshop safety, vehicle architecture, safe vehicle lifting **Day 2:** Engine 4-stroke cycle, basic hand tools mastery **Day 3:** Fuel/ignition systems, fastener identification, tightening sequences **Day 4:** Intake system understanding, basic measurements **Day 5:** Exhaust system knowledge, component removal technique **Day 6:** Forced induction awareness, professional tool management

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Week 1 Practical Skills Achieved:

✅ **Safety Skills:**

- Uses PPE correctly
- Lifts vehicles safely with jack and stands
- Works safely under vehicles
- Handles hot components properly
- Avoids carbon monoxide hazards

✅ **Basic Mechanical Skills:**

- Removes and installs wheels with proper torque sequence
- Removes battery with correct terminal sequence
- Uses basic hand tools correctly (spanners, ratchets, screwdrivers)
- Selects correct tool for fastener size
- Removes simple bolted components
- Organizes parts during disassembly

- Tracks fasteners to prevent mixing

✅ **Measurement & Documentation:**

- Uses tape measure for basic dimensions
- Operates digital calipers
- Uses torque wrench at basic level
- Documents work with photos
- Creates inspection reports

✅ **System Identification:**

- Identifies major vehicle structural components
- Traces intake air path completely
- Traces exhaust gas path completely
- Locates major engine components
- Identifies forced induction systems

✅ **Professional Habits:**

- Organizes tool box systematically
- Maintains tools properly
- Cleans work area
- Follows proper sequences
- Works methodically, not rushing

Week 1 Knowledge Gained:

Engine Systems:

- 4-stroke cycle operation
- Major engine components and functions
- Fuel system path and components
- Ignition system components
- Timing concepts
- Intake and exhaust flow
- Forced induction principles

Safety Knowledge:

- Workshop hazards and prevention
- PPE selection and use

- Vehicle lifting safety
- Carbon monoxide danger
- Electrical safety (battery terminals)
- Heat hazards (exhaust)

Professional Knowledge:

- Tool selection and usage
- Fastener identification
- Torque specifications
- Measurement principles
- Documentation importance
- Parts organization
- Tool maintenance

Week 1 Assessment Methods:

Daily Assessments:

- Practical skills checklists (completed each day)
- Verbal knowledge checks (instructor questions)
- Hands-on demonstrations
- Photo documentation review

Week 1 Final Assessment (Day 6 afternoon - can be added):

Practical Exam (90 minutes):

1. Vehicle lifting and wheel removal (15 min)
2. Tool selection challenge (10 min)
3. Component identification test (15 min)
4. Battery removal/installation (10 min)
5. Air filter inspection (10 min)
6. Torque wrench usage test (10 min)
7. Parts organization exercise (10 min)
8. Tool maintenance demonstration (10 min)

Written Exam (60 minutes):

- 25 multiple choice questions
- 10 diagram labeling questions
- 5 short answer questions

- Topics: Safety, engine basics, systems, tools, procedures

Pass Criteria:

- Practical: 70% minimum (perform skills correctly)
- Written: 60% minimum (understand concepts)
- Safety: 100% (no safety violations allowed)